Why are depression symptoms scores greater in people experiencing romantic love?

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1 Author contributions

AB conceived of the model outlined in the article, conducted the Fisher's exact tests and calculated effect sizes, and wrote the article. LK provided the data and undertook the statistical analysis related to the Mann-Whitney U tests and provided comments on earlier version of the article. All authors approved the final article for publication.

2 Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Abstract

This article presents a model to explain why individuals experiencing romantic love have greater depression symptoms scores. We start by defining romantic love and providing a narrative review of studies investigating depression symptoms in people experiencing romantic love. We then present data from Finnish adolescents which shows higher depression symptoms scores in groups experiencing romantic love. Some tests indicate a small effect of romantic love on depression symptoms scores. We outline biological mechanisms that may play a role in generating depression symptoms in people experiencing romantic love before proposing a hypothetical model to explain the association. This article concludes with a discussion that provides further information about the causes and functions of depression symptoms in people experiencing romantic love.

1 Introduction

There is evidence that greater depression symptoms scores are associated with romantic love in adolescents (Kuula, Partonen, & Pesonen, 2020) and young adults (Bajoghli et al., 2014; Brand et al., 2015). Mood serves evolutionary functions (see Nettle & Bateson, 2012). Elevated mood can promote goal-oriented behavior and depressed mood can extinguish goal-oriented behavior (Wrosch & Miller, 2009; Bindl, Parker, Totterdell, & Hagger-Johnson, 2012; Nesse, 2019). Different precipitants are associated with different patterns of depression symptoms, possibly because these symptoms serve different functions (Keller & Nesse, 2005, 2006). Some types of rumination may help individuals focus on, and resolve, complex social problems (Andrews & Thomson, 2009).

This article proposes a model to explain increased depression symptoms scores in people experiencing romantic love. First, romantic love is defined. Second, a summary of the research investigating depression symptoms in adolescents and young adults experiencing romantic love is presented. Third, we introduce data from Finnish adolescents to support the claim that depression symptoms scores are higher in people experiencing romantic love. Fourth, we summarize the mechanisms that may contribute to a relationship between depression symptoms and romantic love. Fifth, we propose a model to explain the increased depression symptoms scores in people experiencing romantic love. This article concludes with a discussion. We suggest that depression symptoms may have different catalysts and serve different functions in people experiencing romantic love, discuss whether depression symptoms in people experiencing romantic love are adaptations or by-products, and identify areas for future research.

2 Romantic love

Romantic love is:

> … a motivational state typically associated with a desire for long-term mating with a particular individual. It occurs across the lifespan and is associated with distinctive cognitive, emotional, behavioral, social, genetic, neural, and endocrine activity in both sexes. Throughout much of the life course, it serves mate choice, courtship, sex, and pair-bonding functions. It is a suite of adaptations and by-products that arose sometime during the recent evolutionary history of humans (Bode & Kushnick, 2021, p. 21).

Romantic love can develop in childhood (Hatfield, Schmitz, Cornelius, & Rapson, 1988) but is common in, and most commonly associated with, adolescence (Hatfield & Sprecher, 1986; Hill, Blakemore, & Drumm, 1997). It is common at the beginning of a romantic relationship but can occur in the absence of a romantic relationship. The mechanisms that cause romantic love, its development across the lifespan, its functions, and its evolutionary history have been described in detail elsewhere (Bode & Kushnick, 2021). That review suggested that romantic love has neurobiological and endocrinological similarities with depression.

3 Depression symptoms and romantic love

To our knowledge, there have been four studies investigating depression symptoms in adolescents experiencing romantic love (Bajoghli, Joshaghani, Mohammadi, Holsboer-Trachsler, & Brand, 2011; Bajoghli et al., 2013; Bajoghli et al., 2017; Kuula et al., 2020) and results are mixed. Bajoghli et al. (2011) were the first. They found no association between depression symptoms scores and romantic love in female Iranian high school students, although there was a trend towards a negative association. Later, Bajoghli et al. (2013) investigated depression symptoms scores in female and male

Iranian high school students and found a negative association between romantic love and depression symptoms scores. This study was then used as a baseline by Bajoghli et al. (2017) in a longitudinal study that found that entry into romantic love was associated with lower levels of depression symptoms scores compared to individuals who did not fall in love. That study also found that exit out of romantic love was associated with no change in depression symptoms scores compared to individuals who remained in love. These three studies used Von Zerssen's Depression Scale (VZDS; Von Zerssen, 1976; Appendix 1 presents the VZDS) to assess depression symptoms and a single question about experiencing romantic love to determine romantic love status. These findings contrast with the results of Kuula et al. (2020) who found higher levels of depression symptoms scores among Finnish adolescent females and males experiencing romantic love using the Beck Depression Inventory-II (BDI-II; Beck, Steer, & Brown, 1996) and a single question about being in love. In that study, the aim of which was to investigate the relationship between sleep and feelings of romantic love, females in love had significantly higher depression symptoms scores (15.3) than females who were not in love (12.3). Males in love had significantly higher depression symptoms scores (11.1) than males who were not in love (6.5).

To our knowledge, two studies have investigated depression symptoms in young adults experiencing romantic love (Bajoghli et al., 2014; Brand et al., 2015). Bajoghli et al. (2014) found that more intense romantic love was associated with greater depression symptoms scores among Iranian female and male university students. They used the Beck Depression Inventory (BDI; Beck, Erbaugh, Ward, Mock, & Mendelsohn, 1961; Appendix 2 presents the BDI) to assess depression symptoms and a variation of the Yale-Brown Obsessive Compulsive Scale (Y-BOCS; Goodman et al., 1989; Appendix 3 presents the variation of the Y-BOCS) to assess the intensity of romantic love. They found a correlation of .31 between romantic love intensity and depression symptoms among 100 students, 96 of which were in love. It was followed by a study of young adults in Switzerland and Germany by Brand et al. (Brand et al., 2015) that also found that a greater intensity of romantic love was associated with greater depression symptoms scores. That study, too, used the BDI to assess depression symptoms and a variation of the Y-BOCS to assess the intensity of romantic love. They found a correlation of .22 between romantic love intensity and depression symptoms in 844 participants who were in love. The findings are surprising because successful romantic relationships tend to be associated with better health and well-being (see Fletcher, Simpson, Campbell, & Overall, 2015), including among young adults (Gomez-Lopez, Viejo, & Ortega-Ruiz, 2019).

3.1 Accounting for inconsistent findings

At first examination, the research into the association between depression symptoms scores and romantic love presents as a conundrum. Research into adolescents found no association, a negative association, and a positive association. Studies of young adults found two positive associations between depression symptoms scores and intensity of romantic love. This is contrary to expectations because romantic relationships among older people are less likely to be associated with depression symptoms scores (see Davila, 2008; Gomez-Lopez et al., 2019). However, greater depression symptoms scores are associated with romantic love in some studies (Bajoghli et al., 2014; Brand et al., 2015; Kuula et al., 2020).

The inconsistent findings among adolescents could be accounted for by the use of different measures of depression. Three studies (Bajoghli et al., 2011; Bajoghli et al., 2013; Bajoghli et al., 2017) that found no, or a negative association, used the VZDS and one study that found a positive association (Kuula et al., 2020) used the BDI-II. The BDI-II is a widely used and well validated measure of depression (Wang & Gorenstein, 2013) that aligns with DSM-IV criteria. It appears to be a much

more useful tool to identify sub-clinical depression symptoms than the VZDS. In our opinion, the different measures of depression symptoms are the most likely cause for the inconsistent findings in adolescents. However, another factor that could account for the differences is the cultural differences between Iran and Finland. Romantic love is similar cross-culturally but some differences do exist (Feybesse & Hatfield, 2019; Karandashev, 2019). For example, culture has been shown to influence self-reported importance of personality, physical appearance, propinquity, similarity, readiness, isolation, mystery, and social standing in the development of romantic love (Sprecher et al., 1994).

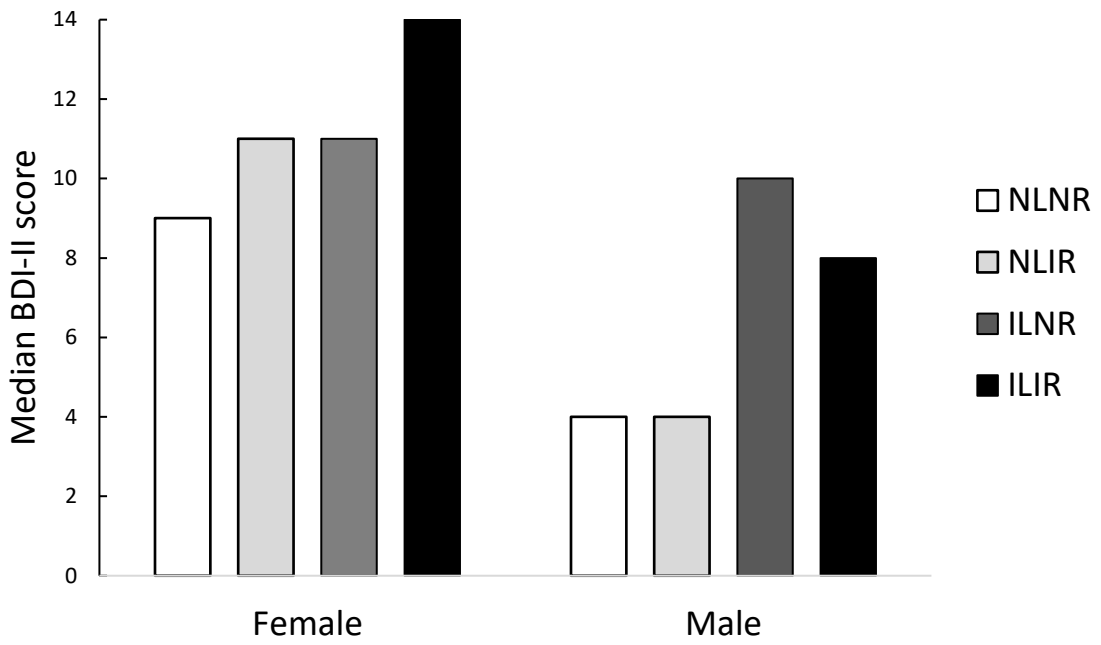
The findings that the intensity of romantic love is associated with depression symptoms scores in young adults may owe itself to the measures of depression symptoms and romantic love in relevant studies. Both studies (Bajoghli et al., 2014; Brand et al., 2015) used the BDI to assess depression symptoms and a variation of the Y-BOCS to assess romantic love. The BDI is a robust measure and the use of the variation of the Y-BOCS provided a measure of romantic love that was sufficiently sensitive to determine intensity and generate a continuous variable rather than relying on a dichotomous variable. This is important because people can be mistaken about whether they are experiencing romantic love (see Campbell, Hosseini, Myers, & Calub, 2016).

4 Depression symptoms in specific groups of Finnish adolescents

The following data analysis is a follow-up to a previous study conducted by Kuula, Partonen, & Pesonen (2020). A detailed description of the sample and measures are provided there. In this analysis, we draw on data from 1,358 participants (girls = 66.3%) to the SleepHelsinki! study, born in 1999 or 2000, who responded with data about depression symptoms, love status, and relationship status. All participants were 16 or 17 years old at the time of data collection. Love status was determined by asking participants the following: “Love can be a powerful feeling, especially in the beginning. Are you currently experiencing the emotional turmoil which relates to the early stages of falling in love?” Relationship status was determined by asking participants the following: “Are you in an exclusive romantic relationship?” Ethical permission was obtained from The Hospital District of Helsinki and Uusimaa Ethics Committee for gynecology and obstetrics, pediatrics, and psychiatry (Decision number 50/13/ 03/03/2016).

Figure 1 presents median BDI-II scores for 4 groups of Finnish adolescents according to sex, love status, and relationship status (not all reported in the original study). These groups are not in love and not in a relationship (NLNR), not in love but in a relationship (NLIR), in love but not in a relationship (ILNR), and in love and in a relationship (ILIR). We undertook Mann-Whitney U tests to assess if BDI-II scores differed between relevant groups. BDI-II scores differed between female NLNR and female ILNR groups (U = 13999, p = .070, r = .070). BDI-II scores differed between female NLIR and female ILIR groups (U = 2294, p = .268, r = .154). BD-II scores differed between male NLNR and male ILNR groups (U = 3126, p < .001, r = .176). This result should be interpreted with caution, however, because the assumption of equal variance between these groups was not met ($F_{1,357}$ = 7.77, p = .006). This increases the risk of type I error in significance tests (Kasuya, 2001). BDI-II scores differed between male NLIR and male ILIR groups (U = 190, p = .272, r = .144).

Figure 1. Median BDI-II scores for four groups of Finnish adolescents by sex



NLNR=Not in love and not in a relationship (female n=597; male n=329); NLIR=Not in love but in a relationship (female n=149; male n=44); ILNR=In love but not in a relationship (female n=55; male n=30); ILIR=In love and in a relationship (female n=35; male n=11)

Table 1 presents the proportion of each group with scores on the BDI-II that indicate no depression symptoms, mild range, mild depression, moderate depression, and severe depression (see Beck et al., 1996). Table 1 presents the percentage of each group meeting thresholds for levels of depression. We recognize that the prevalence of scores meeting thresholds for clinical depression appears high in this sample. BD-II threshold percentages did not differ between female NILNR and female ILNR groups (p = 0.212, 2-sided Fisher's exact test). BDI-II threshold percentages did not differ between female NLIR and female ILIR groups (p = 0.298, 2-sided Fisher's exact test). BDI-II threshold percentages differed between male NLNR and male ILNR groups (p = 0.002, 2-sided Fisher's exact test). BDI-II threshold percentages did not differ between male NILIR and male ILIR groups (p = 0.772, 2-sided Fisher's exact test).

5 Biological mechanisms that potentially contribute to depression symptoms in people experiencing romantic love

5.1 Neural and endocrinological activity

Given that there is some evidence that depression symptoms scores are associated with romantic love (including among adolescents), it raises the question of why this is the case. Similar or related biological mechanisms may be one explanation (see Bode & Kushnick, 2021). Both depression and romantic love are associated with changes in functional connectivity in networks involving the amygdala, medial prefrontal cortex, and nucleus accumbens (Dean & Keshavan, 2017; Song et al., 2015). There are also endocrinological similarities, including activity in the monoamine (Dean & Keshavan, 2017), cortisol (Dean & Keshavan, 2017), and oxytocin (McQuaid, McInnis, Abizaid, & Anisman, 2014; Neumann & Landgraf, 2012; Purba, Hoogendijk, Hofman, & Swaab, 1996; Van Londen et al., 1998) systems (see Bode & Kushnick, 2021 for review).

5.2 The behavioral inhibition system (BIS)

Mood is an emotional predictor of the prospects of pleasure and pain (Morris, 2003). Elevated mood promotes goal-oriented behavior and depressed mood extinguishes goal-oriented behavior (Wrosch & Miller, 2009; Bindl et al., 2012; Nesse, 2019). Two distinct mechanisms are believed to play a role in recruiting behavior towards or away from a goal. The behavioral activation system and the behavioral inhibition system (BIS; see Carver & White, 1994) promote and extinguish goal-oriented behavior, respectively. These two mechanisms evolved to integrate information about the environment and the state of the individual to focus decisions about behavioral effort (Nettle & Bateson, 2012). They operate in a manner whereby inputs (stimuli from the environment and state of the individual) are processed, and depending on the sensitivity of the systems, various outputs are produced. Outputs of the BIS manifest as depression symptoms, with varying degrees of intensity. It may involve the amygdala (Morgan, 2006), right orbitofrontal cortex, and precuneus (Fuentes et al., 2012). In romantic love, the behavioral activation system is probably engaged, manifesting in elevated mood (Brand, Luethi, von Planta, Hatzinger, & Holsboer-Trachsler, 2007; Bajoghli et al., 2011; Bajoghli et al., 2013; Bajoghli et al., 2014; Brand et al., 2015). The goals are to facilitate courtship, sex, and pair-bonding with a loved one (see Bode & Kushnick, 2021). The association between romantic love and depression symptoms scores suggests that the BIS may also be engaged, promoting symptoms of depression.

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Table 1. Percentage of four groups of Finnish adolscents meeting thresholds for depression by sex

	Females				Males			
	NLNR	**NLIR**	**ILNR**	**ILIR**	**NLNR**	**NLIR**	**ILNR**	**ILIR**
No depression symptoms (0)	6.5	2.0	5.5	5.7	17.3	13.6	3.3	9.1
Minimal range (1-13)	59.7	54.4	52.7	42.9	70.5	61.4	60.0	54.5
Mild depression (14-19)	13.9	17.4	9.1	20.0	7.3	13.6	16.6	18.2
Moderate depression (20-28)	11.4	19.5	21.8	17.1	3.4	2.3	16.6	9.1
Severe depression (29-63)	8.5	6.7	10.9	14.3	1.5	9.1	3.3	9.1

NLNR=Not in love and not in a relationship (female n=597; male n=329); NLIR=Not in love but in a relationship (female n=149; male n=44); ILNR=In love but not in a relationship (female n=55; male n=30); ILIR=In love and in a relationship (female n=35; male n=11)

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5.3 The sustained analysis system (SAS)

One theory of the adaptive function of rumination in depressed mood is that it is a response to complex problems, and its function is to minimize disruption and sustain analysis of those problems (Andrews & Thomson, 2009). Others frame rumination in terms of a focus on events that may interfere with goal achievement to promote effective goal-pursuit (Kennair, Haarklau Kleppestø, Larsen, & Jørgensen, 2017; Trincas, Schepisi, Leombruni, Di Mauro, & Mancini, 2018). These approaches are complementary. There have been some studies that lend support for the adaptive function of rumination (Maslej, Mulsant, & Andrews, 2020; Ševčíková et al., 2020). In our opinion, it seems probable that humans would possess a system or systems that focus attention on complex social problems. We are noncommittal, however, on whether rumination is a system acting in an adaptative manner or whether it is a consequence of a malfunction of that system.

A sustained analysis system (SAS) may give a triggering problem prioritized access to processing resources, and is associated with a reduced desire to engage in distracting activities and psychomotor changes that reduce exposure to distracting stimuli (Andrews & Thomson, 2009). There are fewer social problems that are more complex than the formation of a pair-bond and the creation of offspring. Such challenges require the reassessment of priorities, a reorganizing of behavioral effort (Fisher, 1998), and long-term commitment to the well-being of bonded unions and their members. The SAS generates its own particular outputs, including rumination, to promote sustained analysis of social and other challenges. Rumination often precedes the onset of depressive symptomology (see Nolen-Hoeksema, Wisco, & Lyubomirsky, 2008; Wilkinson, Croudace, & Goodyer, 2013) and is associated with more severe depression symptoms in people with unipolar depression (Kuehner & Weber, 1999). This suggests that rumination may have a causal effect on depression symptoms in addition to potential exacerbating and maintaining effects. The SAS, in the context of depressed mood, involves the left ventrolateral prefrontal cortex (LVLPC) and the serotonin system acting on working memory (Andrews & Thomson, 2009).

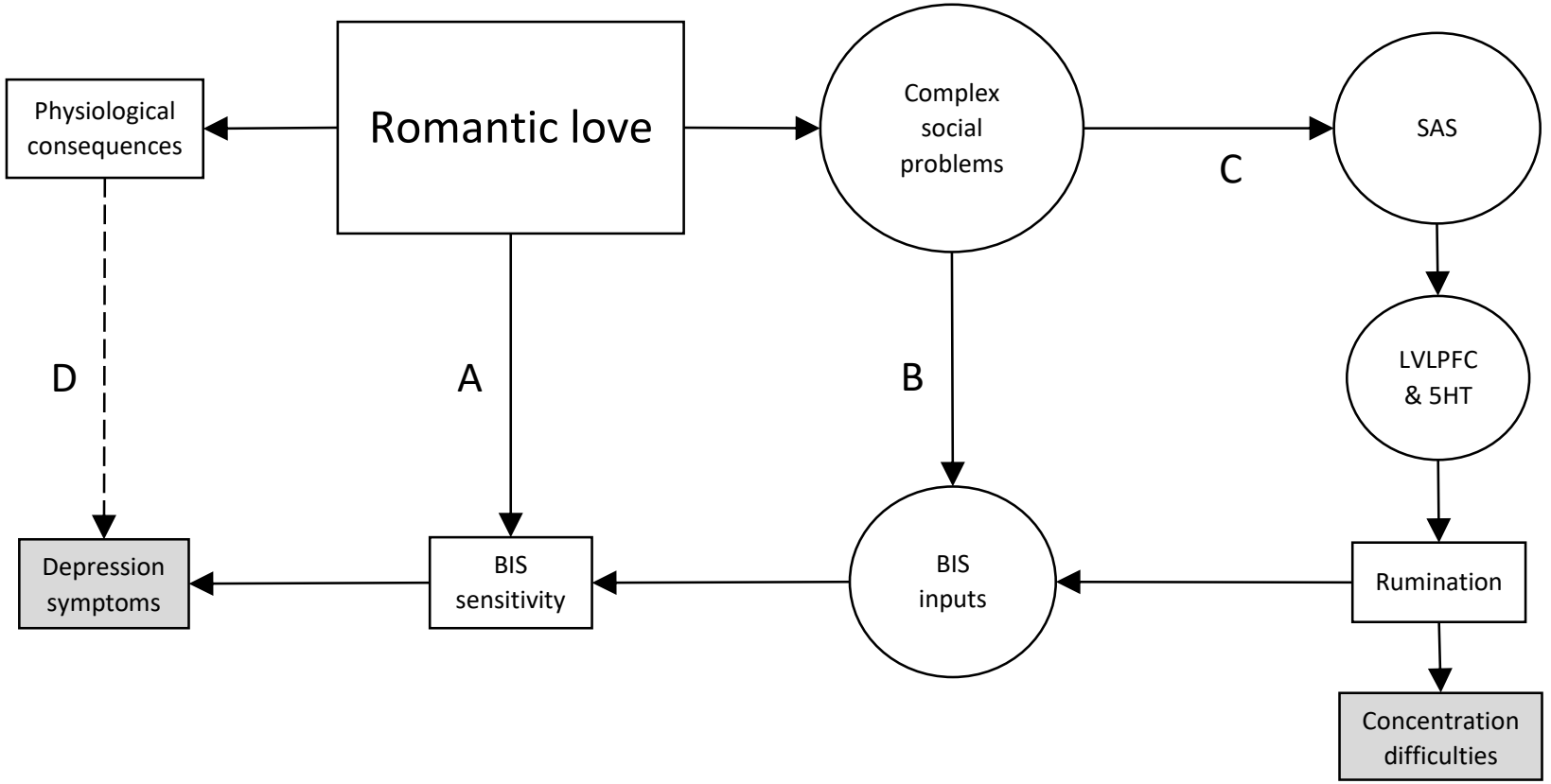
6 A model explaining how romantic love may cause increased depression symptoms scores

Given the two evolutionary-informed theories of depression mood involving the BIS and SAS, it is possible for us to propose a model explaining how romantic love may cause increased depression symptoms scores. Figure 2 presents this model. Included in the model are romantic love, complex social problems, the behavioral inhibition system, the sustained analysis system, and the outputs of these systems (depression symptoms and rumination, respectively). Romantic love may cause depression symptoms in four ways: (A) increasing BIS sensitivity, (B) complex social problems provide BIS inputs, (C) complex social problems activate the sustained analysis system to cause rumination which results in concentration difficulties and provides further BIS inputs, and (D) physiological characteristics of romantic love that mimic depression symptoms.

6.1 Increased BIS sensitivity (A)

The first proposed pathway that may account for increased depression symptoms scores in people experiencing romantic love is one where romantic love acts directly on the BIS by increasing sensitivity to inputs related to the loved one (pathway A in Figure 2). This means that what would normally be relatively innocuous stimuli elicit a stronger BIS response, leading to greater depression symptoms. This sensitivity probably applies only to stimuli related to the loved one. Negative feelings when things go awry with their loved ones are commonly reported among people experiencing romantic love (Hatfield & Sprecher, 1986). This increased BIS sensitivity would, thus,

Figure 2. Hypothetical model explaining how romantic love may cause increased depression symptoms scores



Notes. Latent constructs are circles. Manifest variables that can be measured with survey instruments are rectangles. Gray rectangles represent depression symptoms. Romantic love causes depression through four pathways: (A) increasing BIS sensitivity, (B) complex social problems provide BIS inputs, (C) complex social problems activate the SAS which indirectly causes depression symptoms through rumination, and (D) physiological consequences mimicking depression symptoms. The dashed line represents a pathway that is not indicative of depressed mood. Rumination causes concentration difficulties and causes BIS inputs. BIS=Behavioral inhibition system; SAS=Sustained analysis system; LVLPFC=Left ventrolateral prefrontal cortex; 5HT=Serotonin.

differ from the increased BIS sensitivity associated with major depressive disorder (see Arfaie, Safikhanlou, Bakhshipour Roodsari, Farnam, & Shafiee-Kandjani, 2018).

6.2 Complex social problems provide BIS inputs (B)

A second proposed pathway that may account for increased depression symptoms scores in people experiencing romantic love involves the development of complex social problems. This social problem could relate to mate choice, courtship, sex, or pair-bonding, or could be something only peripherally related to romantic love but which is exacerbated by the social situation associated with romantic love. In the case of females, in our evolutionary history, pregnancy would also have been a complex social problem created by romantic love (Bode & Kushnick, 2021). These complex social problems include stimuli that are perceived as negative and serve as a BIS input (pathway B in Figure 2). The downstream consequences of these increased inputs would be increased depression symptoms (see Andrews & Thomson, 2009).

6.3 Complex social problems activate the SAS (C)

A third proposed pathway that may account for increased depression symptoms scores in people experiencing romantic love also involves the creation of complex social problems. In this pathway, however, complex social problems activate the SAS which induces rumination (pathway C in Figure 2). Rumination results from activation of neural structures (LVLPC) and neurotransmitter (serotonin) systems influencing working memory (Andrews & Thomson, 2009). This rumination makes the individual experience repeated thoughts of negative or aversive stimuli which generate concentration difficulties. Rumination also acts as BIS inputs, ultimately resulting in further depression symptoms, particularly anhedonia and psychomotor changes. Evidence indicates that rumination often precedes the onset of depression (see Michl, McLaughlin, Shepherd, & Nolen-Hoeksema, 2013; Nolen-Hoeksema et al., 2008; Wilkinson et al., 2013) and is associated with anxiety (Dar & Iqbal, 2015) (Michl et al., 2013; Olatunji, Naragon-Gainey, & Wolitzky-Taylor, 2013). In the context of anxiety, it may be associated with activation of the right ventrolateral prefrontal cortex, rather than the LVLPC (Andrews & Thomson, 2009). This suggests complex social problems may activate the SAS without first activating the BIS. This effect may be exacerbated by romantic love's prioritization of cognitive resources towards loved one-related stimuli (see Langeslag, Olivier, Köhlen, Nijs, & Van Strien, 2015).

6.4 Physiological consequences of romantic love mimic depression symptoms (D)

A fourth and final pathway through which romantic love may cause depression symptoms scores is through physiological consequences of romantic love that mimic depression symptoms (pathway D in Figure 2). Obsessive thinking and crying are two outputs that may mimic depression symptoms in people experiencing romantic love. Obsessive thinking is characteristic of romantic love (Hatfield & Sprecher, 1986; Fisher, 1998) and these thoughts may result in concentration difficulties. One of the questions used to determine the intensity of romantic love in studies assessing romantic love and depression symptoms (Bajoghli et al., 2014; Brand et al., 2015) was "While thinking about the loved person, do you feel distracted?". Additionally, crying can occur in moments of happiness (Avery, 1983; Feldman, 1956; Weiss, 1952)) or during or following sex (Burri & Spector, 2011) which are increased in people experiencing romantic love with their partner. Romantic love may also negatively interfere with sleep (see Bode & Kuula, 2021 for review), which can result in people endorsing depression symptoms in some measures of depression (e.g., BDI, BDI-II), such as sleep changes, tiredness, or fatigue.

7 Discussion

In the sections above, we defined romantic love, summarized the evidence for an association between depression symptoms and romantic love, presented data about depression symptoms for groups of Finnish adolescents, some of whom self-reported experiencing romantic love, introduced the biological mechanisms potentially contributing to depression symptoms in people experiencing romantic love, and proposed a hypothetical model to explain elevated depression symptoms scores in people experiencing romantic love. In this discussion, we highlight that the association between romantic love and depression symptoms scores is small and suggest that depression symptoms may have different catalysts and serve different functions in people experiencing romantic love. We also consider whether depression symptoms in people experiencing romantic love might be adaptations or by-products and identify areas for future research.

7.1 The association between romantic love and depression symptoms scores is small

It is important not to overstate the relationship between romantic love and depression symptoms scores. The effect sizes in relevant studies are relatively small (see Schäfer & Schwarz, 2019) for summary of effect sizes). The effect size for the Mann-Whitny U tests conducted by us ranged from very small (.070) to small (.176). Additionally, the correlations between the intensity of romantic love and BDI scores in Bajoghli et al.'s (2014) and Brand et al.'s (2015) studies amount to only moderate (r = .31) and small (r = .22) effect sizes, respectively. Table 2 presents the effect size and related statistics for the Mann-Whitney U tests presented in this article.

7.2 Depression symptoms may have different catalysts and serve different functions in adolescents

Depression has been associated with adolescent romantic relationships, especially among females (see Davila, 2008 for a review of the early literature), although the evidence is mixed (Gomez-Lopez et al., 2019). Four theories that have been proposed to explain the association between adolescent romantic relationships and depression symptoms include the normative trajectory model, attention impairment model, stress and coping model, and individual differences (Davila, 2008). Other factors associated with adolescent romantic relationships have been specifically associated with depression symptoms: violence occurring within the relationship, interracial relationships, sexual activity outside a committed relationship, and imbalance between romantic expectations and reality (see (Davila, 2008 for review). Personal skills and competence to maintain a healthy, satisfying relationship are associated with reduced depression symptoms (Gomez-Lopez et al., 2019).

Despite these risk factors for depression symptoms in adolescent females in a relationship, it appears that depression symptoms may be partly explained by the presence of romantic love (Kuula et al., 2020). We found a small effect of romantic love on BDI-II scores between females NLIR and females ILIR. The participants in Bajoghli et al.'s (2014) and Brand et al.'s (2015) studies were almost exclusively in a romantic relationship. Females may experience early-stage romantic love more intensely than males (Cannas Aghedu, Veneziani, Manari, Feybesse, & Bisiacchi, 2018; Meskó, Zsidó, Láng, & Karádi, 2021). Reciprocated romantic love tends to be more intense than unrequited romantic love (such as when an individual is not in a romantic relationship with their loved one (Bringle, Winnick, & Rydell, 2013). A greater intensity of romantic love is associated with greater depression symptoms scores (Bajoghli et al., 2014; Brand et al., 2015). This may account, in part, for the counterintuitive findings that females in love and in a relationship had the highest median BDI-II scores among females in our sample. This relationship may act via the increased BIS sensitivity outlined in our model above (pathway A in Figure 2). The existence of a romantic

Table 2. Effect size and related statistics for comparisons of median BDI-II scores made in this article

	U	*Z*	*n*	*r*	Effect size
Female					
NLNR v ILNR	13,999	-1.811	652	.070	**Very small**
NLIR v ILIR	2,294	-1.107	184	.154	**Small**
Male					
NLNR v ILNR	3,126	-3.338	359	.176	**Small**
NLIR v ILIR	190	-1.099	55	.144	**Small**

NLNR=Not in love and not in a relationship; ILNR=In love and in a relationship; ILNR=In love and not in a relationship; ILIR=In love and in a relationship

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relationship, however, may also represent complex social problems for females in love. Female adolescents may have to navigate establishing a relationship, sex, and relationship conflict (Volpe, Morales-Alemán, & Teitelman, 2014), all of which can result in BIS inputs (pathway B in Figure 2) or SAS activation (pathway C in Figure 2). Rumination has been shown to impair implicit affect regulation in response to negative feelings about a relationship or partner (Jostmann, Karremans, & Finkenauer, 2011).

In our evolutionary history, romantic love may have also generated depression symptoms in females who were in a relationship as a means of refocusing energies from pair-bonding, towards pregnancy and child-rearing. For most of human existence, once a romantic relationship had been formed and the type of sexual activity that is associated with romantic love was occurring, pregnancy would usually follow (see Bode & Kushnick, 2021; Goetz, Pillsworth, Buss, & Conroy-Beam, 2019). Pregnancy represents a type of complex social problem which could contribute to rumination and worry in some people (see Kalmbach et al., 2020; Schmidt, Seehagen, Vocks, Schneider, & Teismann, 2016).

Individuals in love but not in a relationship are experiencing what the literature refers to as unrequited love. Unrequited love can take multiple forms, including love for a celebrity, love for someone that an individual is not courting, love for someone an individual is courting, and love for a past lover (Bringle et al., 2013). Unrequited love is more common among males than females (see (Hill et al., 1997; Kuula et al., 2020), and different types of unrequited love are associated with different durations of romantic love (Bringle et al., 2013).

The theory of parental investment (Trivers, 1972) states that the sex that invests more into reproduction will be choosier in terms of selecting mates. In humans, females become pregnant and then care for the infant during a period of lactation. As a result, they are choosier than men in terms of selecting mates (although males are also choosy when selecting long-term mates; see Stewart-Williams & Thomas, 2013). Because of this choosiness, males, to a greater extent, engage in a process of active courtship to prove themselves worthy long-term mates to females. They may be more likely to experience long-term mate choice (i.e., romantic love) and associated efforts to initiate a romantic relationship (Kuula et al., 2020; Hill et al., 1997). Because males are more likely to experience unrequited love than females, there is a greater frequency of need to extinguish feelings of romantic love in males when pursuit of the loved one will not be fruitful.

Unrequited love for someone an individual is courting and love for a past lover, in particular, are associated with reward pursuit. If efforts to initiate, maintain, or re-instigate a romantic relationship are unsuccessful, an individual is exposed to a negative outcome in the form of a complex social problem. These would be associated with both BIS and SAS activity (pathways B and C in Figure 2). Here we suggest that depression symptoms associated with males being in love but not in a relationship may cumulatively serve the adaptive function of inhibiting courtship behaviors, prioritizing analysis of the complex social problem of unrequited love, and promoting the extinction of romantic love. This conserves energy and redirects reproductive efforts elsewhere.

7.3 Depression symptoms associated with romantic relationship dissolution

One pathway omitted from our model is that in which romantic relationship dissolution causes clinical depression or depression symptoms. One type of unrequited love is love for a former partner (Bringle et al., 2013). This would constitute a proportion of Finnish adolescents in love but not in a relationship in our data. Romantic relationship dissolution (i.e., a breakup) has been associated with negative mood (e.g., Keller & Nesse, 2005), depression symptoms (e.g., Verhallen, Renken,

Marsman, & ter Horst, 2019), and clinical depression (e.g., Mearns, 1991; Monroe, Rohde, Seeley, & Lewinsohn, 1999). It can best be considered in terms of being one type of pair-bond disruption. A review (Lieberwirth & Wang, 2016) summarizes the literature on pair-bond disruption in monogamous prairie voles. The evidence indicates that separation from a bonded partner elevates plasma corticosterone and increases anxiety- and depression-like behaviors. The pair-bond, measured by partner preferences, was extinguished after four weeks of separation. These findings indicate a role of the hypothalamic-pituitary-adrenal axis in pair-bond disruption and that pair-bond disruption is causally related to depression-like behaviors (Lieberwirth & Wang, 2016). In humans, romantic relationship dissolution is associated with similar outcomes which are distinct from other types of pair-bond disruptions, such as the loss of a partner to death (Burger et al., 2020). Whether a breakup is wanted or unwanted is an obvious moderator of these effects, although negative outcomes are associated with romantic relationship dissolution generally.

Social mechanisms (e.g., Gähler, 2006; Biotteau, Bonnet, & Cambois, 2019), psychological mechanisms (e.g., Boelen & Reijntjes, 2009; van Tilburg, Aartsen, & van der Pas, 2015), neurobiological mechanisms (see van de Watt et a., 2021; see also Stoessel et al., 2011; Song et al., 2015; Alonso Martínez, Marsman, Kringelbach, Deco, & ter Horst, 2020) and (neuro)endocrinological mechanisms (see Fisher, Xu, Aron, & Brown, 2016) play a role in depression symptoms in people who have experienced romantic relationship dissolution. It is important to recognize, as well, studies of depression symptoms following romantic relationship dissolution have not tended to investigate the specific context of romantic love, which usually only lasts an average of under two years (Bringle et al., 2013). As a result, we believe it is credible to say that the depression symptoms following romantic relationship dissolution are the result of pair-bond disruption, rather than romantic love.

Romantic love serves a pair-bonding function, and therefore serves to creates attachment, the disruption of which causes the depression symptoms. The "withdrawal"-like consequences of romantic relationship dissolution (see Fisher et al., 2016) occur whether an individual is experiencing romantic love or in an established pair-bond. We believe that opioids (i.e. beta-endorphin; see Machin & Dunbar, 2011) and their interaction with the dopamine system are probably a driving force behind the withdrawal-type effects of dissolution, rather than simply dopamine. Dopamine, does however, play a downstream role (see Diana, Muntoni, Pistis, Melis, & Gessa, 1999) and, potentially, a primary role in individuals who are experiencing romantic love at the time of dissolution. As such, while the issue of depression symptoms associated with romantic relationship dissolution is pertinent and may partially explain increased depression scores in Finnish adolescents in love but not in a relationship compared with adolescents not in love and not in a relationship, we feel justified in not including it in our model. We do, however, believe that differences in depression symptoms following romantic relationship dissolution between individuals experiencing romantic love and those not experiencing romantic love is a valid area for future research.

7.4 Does attachment style influence the relationship between romantic love and depression symptoms scores?

Attachment is one aspect of close social bonds (see Fraley & Shaver, 2000). It is a psychological manifestation of a biological system that serves multiple evolutionary functions (see Barbaro, 2020). It is associated with adult pair-bonds (Zeifman & Hazan, 2008) and different attachment styles are associated with different cognitions, emotions, and behaviors (e.g., Kim & Miller, 2020; Sheinbaum et al., 2015) as well as different risks for psychopathology. (see Mikulincer & Shaver, 2012). As

such, it is reasonable to ask the question, does attachment style influence the relationship between romantic love and depression symptoms scores.

Anxious attachment styles are associated with anxiety and vigilance concerning rejection and abandonment (Brennan, Clark, & Shaver, 1998). Individuals with this attachment style may experience elevated BIS sensitivity to inputs associated with rejection by a potential partner, risk of abandonment by an existing partner, or abandonment by a former partner. Bad feelings when things go wrong is a feature of romantic love (Hatfield & Sprecher, 1986) and this may be exacerbated in people with an anxious attachment style. Additionally, there may be a tendency to overperceive these sorts of stimuli compared to others, which simply results in increased BIS inputs. To this extent, anxious attachment style may result in increased depression symptoms among individuals who are not in a relationship with their loved one as well as individuals who are in a relationship with their loved one.

Avoidant attachment styles are associated with discomfort with closeness and dependency or reluctance to be intimate with others (Brennan, Clark, & Shaver, 1998). Individuals with this attachment style may interpret aspects of courtship or romantic relationships, such as obtrusive thoughts or commitment, as aversive, meaning they are receiving increased BIS inputs. To this extent, avoidant attachment style may result in increased depression symptoms among individuals who are not in a relationship with their loved one as well as individuals who are in a relationship with their loved one.

The attachment system plays a role in, but is not completely, part of romantic love. Romantic love serves a pair-bonding function (Bode & Kushnick, 2021) and pair-bonding is the biological process that creates adult romantic attachment. So, while attachment styles may be particularly relevant to depression symptoms in people experiencing romantic love, they do not belong in our model. Instead, research should attempt to identify any role attachment style might play in increasing depression symptoms in people experiencing romantic love, either as a moderator or confounder (see Simpson & Steven Rholes, 2017 for ideas).

7.5 Does life history strategy influence the relationship between romantic love and depression symptoms scores?

Life history theory, applied to variability among humans, is a theory inspired by evolutionary biological theory and research that demonstrated that species possess different reproductive, metabolic, and growth characteristics (see Sear, 2020 for description of life history theories; see also Hill, 1993; Nettle & Frankenhuis, 2019). This framework has been applied to humans in the context of allocating a "life history strategy" to different individuals. These strategies are categorized based upon location on a fast-slow continuum. Individuals that exhibit a "fast" life history strategy display relatively faster growth, reproduce relatively more frequently and at an earlier age, and are prone to more rapid senescence and earlier death (Sear, 2020). These strategies have been associated with particular cognitive and behavioral expression and those characteristics can be referred to as psychological life-history (Sear, 2020).

Each life history strategy may be associated with the different expression of romantic love. A loving style similar to romantic love (eros; see Hendrick & Hendrick, 1986) tends to be associated more with a slow life history strategy than other love styles (Marzec and Łukasik, 2017). Genetic determinants of life history strategy may influence the occurrence of romantic love (see Bode & Kushnick, 2021). Life history strategies may also be differentially associated with different psychopathologies, including particular subtypes of depression (see Belsky, 2014; Del Giudice, 2014;

Del Giudice, 2015). To that extent, life history strategy may influence the relationship between romantic love and depression symptoms. At the very least, future research should consider if life history strategy moderates or confounds that relationship.

7.6 Depressed mood versus clinical depression

Our proposed model tries to account for increased depression symptoms scores in people experiencing romantic love. As a result, it is important to clarify the distinction between depressed mood and clinical depression. Depressed mood is characterized by milder or fewer depression symptoms than clinical depression. Depression symptoms in people experiencing romantic love may increase to levels indicative of clinical depression but not be a psychopathology. While some people experiencing romantic love may have clinical depression prior to the onset of romantic love, the increase in depression symptoms among those experiencing romantic love may not represent symptoms of clinical depression. We can't, however, discount the possibility that romantic love may trigger or exacerbate clinical depression. Romantic love has been associated with the onset of psychopathology (McLauchlan et al., 2021). To that extent, the hypotheses contained in this article justify consideration by psychiatric and psychological clinicians.

The social navigation hypothesis (Watson & Andrews, 2002) posits that clinical depression can serve to focus attention on social problems and motivate members of the depressive's social network to make investments or concessions. To that extent, it includes aspects of the sustained analysis system outlined in our model. We reserve judgment on whether clinical depression is adaptive, although we note that there is a tendency upon clinical evolutionary psychologists to suggest otherwise (Kennair et al., 2017). Advocates of the social navigation hypothesis (Cline-Brown & Watson, 2005) believe that mild or moderate clinical depression can facilitate the sustained analysis of complex problems. This suggests that lower levels of depression symptoms, such as those that may be found in people experiencing romantic love, may serve the evolutionary functions suggested by our model. Other evolutionary theorists contend that sub-clinical depression symptoms (i.e., normal low mood) serve adaptive functions that align with the behavioral inhibition component of our model (Nesse, 2019; Nettle & Bateson, 2012).

7.7 Romantic love is a suite of adaptations and by-products

Romantic love is a suite of adaptations and by-products (Bode & Kushnick, 2021; Buss, 2019). Depression symptoms in people experiencing romantic love may be both an adaptation and by-product. An adaptation is "…an inherited and reliably developing characteristic that came into existence as a feature of a species through natural selection because it helped to directly or indirectly facilitate reproduction during the period of its evolution" (Buss, Haselton, Shackelford, Bleske, & Wakefield, 1998, p. 535; see also Williams, 2019). Another way of defining adaptation is "a phenotypic variant that results in the highest fitness among a specified set of variants in a given environment" (Reeve & Sherman, 1993, p. 9). Determining if depression symptoms are an adaptation requires consideration of complexity, efficiency, reliability, specificity, capability for solving adaptive problems, evolvability (Buss et al., 1998), and reproductive success (Reeve & Sherman, 1993). However, we can postulate that the process of natural selection may have favored individuals who displayed increased BIS sensitivity towards stimuli associated with the loved one. The reproductive benefit could have been to ensure that challenges to courtship and pair-bonding were given increased weight in the minds of individuals experiencing romantic love. Males, in love, but not in a relationship with their loved one, may be displaying an adaptive depressed mood. We can't discount the possibility, however, that increased BIS activity is a by-product of other mechanisms.

A by-product is a trait that evolved "not because it was selectively advantageous, but because it was inextricably linked […] to another trait that was reproductively advantageous" (Andrews, Gangestad, & Matthews, 2002, p48.). The complex social situation associated with courtship, sex, pair-bonding, and in the case of females in our evolutionary history, pregnancy, may result in BIS inputs and activation of the SAS, not because of any mechanistic changes resulting from romantic love, but simply in response to the situation that arises (pathway B in Figure 2). This would make depression symptoms and rumination in people experiencing romantic love by-products. As a result, depression symptoms in people experiencing romantic love, if they are not caused by romantic love's physiological characteristics mimicking depression symptoms, may be both an adaptation and by-product.

7.8 Limitations of our data and our model

There are a number of limitations of both our data and our model that should be taken into consideration. First, our data is sourced from a single city not representative of the global population. It is important not to overestimate the generalizability of findings to all adolescents (see Apicella, Norenzayan, & Henrich, 2019; Henrich, Heine, & Norenzayan, 2010). There are also issues with the way romantic love was measured. Romantic love was assessed by self report in response to the following question, "Are you currently experiencing the emotional turmoil which relates to the early stages of falling in love? The reference to "emotional turmoil" could have biased the sample and inflated the depression symptoms scores in adolescents self-reporting romantic love. We also note that shorter sleep duration is associated with romantic love in adolescents, but not young adults (Bode & Kuula 2021). As a result, our data, which used an adolescent sample, may be more likely to have romantic lovers endorsing symptoms of depression associated with sleep disturbance, fatigue, and tiredness, than may be the case in older samples.

Our samples were also small, which limited the ability of our analysis to identify significant differences in depression symptoms scores between a number of groups. Small sample sizes result in greater chance of type II error (Lakens, 2022). This means that the visible difference in depression symptoms scores may in fact represent real effects despite not reaching statistical significance (see Wasserstein, Schirm, & Lazar, 2019). It is for this reason that we emphasize effect sizes.

Our model, too, has major limitations. It is hypothetical, and, to our knowledge, none of the pathways that are proposed have been empirically investigated before. Our model proposes roles for systems that are not widely investigated in the literature. There is almost no research investigating what we term the sustained analysis system in the context of adaptive functions. Additionally, given the relatively small effect of romantic love on depression symptoms scores, testing the pathways in the model is going to require substantial sample sizes to maximize statistical power to identify what are likely to be very small effects. Nonetheless, many of the manifest variables that constitute our model can be measured with simple survey instruments, and the usefulness of the model to inform future research is substantial.

7.9 Areas for future research

This article has identified a large number of areas for future research. First, there is a need to extend the research into the association between romantic love and depression symptoms. There is a need to clarify the relationship in, especially, adolescents. Future research should attempt to use empirical measures to assess romantic love status and intensity and robust measures of depression symptoms. It may also be best to start to consider depression symptoms in people experiencing romantic love in terms of negative or depressed mood, rather than depression symptoms.

Second, efforts should be made to test the four pathways identified in our hypothetical model: (A) BIS sensitivity, (B) complex social problems that input into the BIS, (C) complex social problems that activate the SAS, and (D) physiological consequences that mimic depression symptoms. It will be necessary to investigate if BIS sensitivity is increased in people experiencing romantic love only in relation to stimuli related to the loved one and relationship, or whether there is a more global increased BIS sensitivity. Research should attempt to identify and describe the nature of the complex social problem developed by romantic love, in different contexts (e.g., unrequited love or in a relationship). We note that social groups may play a more important role in the lives of adolescents than in young adults or older people. Specific types of problems may be associated with specific depression symptoms. Research should investigate activation of the SAS and downstream consequences. This will require research to investigate rumination in people experiencing romantic love, which may be complicated because of the overlap with obsessive thoughts characteristic of early-stage romantic love. Investigation of specific depression symptoms will also be required (e.g., psychomotor changes, anhedonia, and concentrations difficulties). It will also be necessary to gauge whether concentration difficulties are a result of obsessive thinking or not, and if crying is caused by happiness in some people. The role of sleep variations in people, especially adolescents, experiencing romantic love, should also be investigated as a contributor to increased depression symptom scores (see Bode & Kuula, 2021 for review and discussion).

Finally, it may be necessary to use novel methods to help shed light on the relationship between romantic love and depression symptoms. Consideration of covariates, such as sex, relationship status, and symptoms of psychopathology will be required. We suggest that classifying methods, such as latent class analysis or cluster analysis may provide particularly insightful information that can be used to generate hypotheses and guide future research. We also suggest considering potential covariates, moderators, and confounders, such as attachment style or life history strategy.

8 Conclusion

This article proposed a model to explain increased depression symptoms in people experiencing romantic love. First, romantic love was defined. Second, a summary of the research investigating depression symptoms in adolescents and young adults experiencing romantic love was presented. Third, the mechanisms that may contribute to a relationship between depression symptoms and romantic love were summarized. Fourth, we proposed a model to explain the increased depression symptoms scores in people experiencing romantic love. Fifth, we introduced data from Finnish adolescents and suggest that some depression symptoms in people experiencing romantic love are adaptive. This article concluded with a discussion that suggested that depression symptoms may have different catalysts and serve different functions in people experiencing romantic love, considered whether depression symptoms in people experiencing romantic love are adaptations or by-products, and identified areas for future research.

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